

# Decentralised Knowledge Graph Evolution via Blockchain

Xiangyu Wang<sup>ID</sup>, Taiyu Ban<sup>ID</sup>, Lyuzhou Chen<sup>ID</sup>, Muhammad Usman<sup>ID</sup>, Yifeng Guan<sup>ID</sup>, Derui Lyu<sup>ID</sup>,  
Jian Cheng<sup>ID</sup>, Huanhuan Chen<sup>ID</sup>, *Senior Member, IEEE*, Cyril Leung<sup>ID</sup>, *Life Member, IEEE*,  
and Chunyan Miao<sup>ID</sup>, *Fellow, IEEE*

**Abstract**—In recent years, knowledge graphs (KGs) have been applied in various domains, where the construction and maintenance of the KGs are usually time- and labor-intensive. In this context, constructing shareable KG through multiple constructors is being attempted to reduce costs. In this collaborative process, security and quality issues are critical. The system for constructing shareable KGs should be capable to recover the KG from most malicious attack and to filter out wrong triples from dynamically submitted ones. Blockchain could naturally prevent malicious tampering with its record data, perfect for solving the security issue. However, the integration of multi-source KGs as well as the quality issue still lacks solutions. To address the issues, this paper proposes a blockchain-based high-quality KG collaborative construction framework to ensure the KG quality in its long-term evolution. The framework is built on the underlying consensus mechanism of the blockchain, adopted to an extensible data structure to store multi-source triples on the distributed ledger. A smart contract is implemented to publish triples, assess the contributor credibility and evaluate triple quality to keep the KG in high-quality. Anti-attack mechanisms are designed to defend against malicious triple submissions. Experiments are conducted demonstrating the effectiveness of the framework.

**Index Terms**—Blockchain, knowledge graph, knowledge management, knowledge quality, multi-contributors.

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Xiangyu Wang, Taiyu Ban, Lyuzhou Chen, Muhammad Usman, Yifeng Guan, Derui Lyu, and Huanhuan Chen are with the School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: sa312@mail.ustc.edu.cn; banty@mail.ustc.edu.cn; clz31415@mail.ustc.edu.cn; muhammadusman@mail.ustc.edu.cn; guanyifeng@mail.ustc.edu.cn; drlv@mail.ustc.edu.cn; hchen@ustc.edu.cn).

Jian Cheng is with the Department of Research Institute of Mine Big Data, Chinese Institute of Coal Science, Beijing 100013, China (e-mail: jiancheng@tsinghua.org.cn).

Cyril Leung is with the Department of Electrical and Computer Engineering, The University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: cleung@ece.ubc.ca).

Chunyan Miao is with the School of Computer Science and Engineering, Nanyang Technological University, Singapore 639798 (e-mail: ascymiao@ntu.edu.sg).

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## I. INTRODUCTION

**K**NOWLEDGE Graph (KG), as a special semantic network, represents knowledge<sup>1</sup> by organizing the relationship between entities into triples [1]. With more widespread applications, larger or more domain completed KGs are required to support various tasks [2], [3]. Due to the high labor and time costs for acquiring knowledge, it may be difficult for a small/medium-sized organization to build a KG that meets application requirements [4]. Thus, allowing multiple contributors to build shareable KGs has become an effective way to reduce the cost of KG construction and facilitate its utility. Many large open-sourced KG shared frameworks have been proposed. However, in many tasks, due to the privacy of data, the sharing scope of KGs may be limited, and many existing knowledge sharing frameworks may not be suitable for applications with specific requirements due to different design goals [5]. Therefore, designing a generalized KG construction and sharing framework is of great significance for parties with collaboration and privacy requirements.

For the collaborative construction of shareable KGs, many methods focus on knowledge management, e.g., KG storage, query, visualization, etc. [6], [7]. However, due to the large number of contributors, it is difficult to guarantee the quality of knowledge [8]. In addition, the evolution and improvement of the KG is a long-term process, during which knowledge is added and updated frequently [9], [10]. Therefore, controlling the quality of KGs and avoiding the sharing of low-quality knowledge in this dynamic process is challenging.

First, in multi-contributor knowledge integration, incorrect or inaccurate knowledge inevitably exists in many domains [11]. This may result in contradictions within the KG, especially in the dynamic construction process of the KG. Contradictions may occur not only between the knowledge of different contributors, but also between new knowledge and historical knowledge [12]. Such errors and contradictions greatly affect the quality of KG, and could potentially lead to low quality knowledge if not resolved in a timely fashion.

Second, due to the varied backgrounds of contributors, it is difficult to accurately assess their credibility. In many cases,

<sup>1</sup>Knowledge in this paper refers to the knowledge item in the KG, i.e., triple, which represent knowledge in the form of (*head entity-relation-tail entity*). In many cases, the expression of “triple” pays more attention to its form. In order to facilitate understanding, this paper uses a more general expression, i.e., “knowledge”, to refer to “triples” in many places.

the credibility of contributors may greatly affect the quality of their uploaded knowledge, and contributors with low credibility needs to be carefully managed. Third, on the basis of ensuring the quality of the KG, a strict process is required to meet the maintenance, storage and sharing needs of KG users, while ensuring the privacy and security of data.

These challenges need to be addressed from both knowledge quality (KQ) and knowledge sharing perspectives. For the quality management of KG, a large amount of duplicated/repeated knowledge is uploaded by different contributors in the collaborative construction process. These repeated knowledge could be regarded as consistent descriptions, a kind of evidence for the affirmation of the fact [13], [14]. For example, a description of a fact is considered highly credible when it is consistent with the knowledge provided by a large number of contributors. This kind of consistency information between knowledge does not depend on specific domain information [15]. Therefore, it provides a possible way to evaluate KQ in general cases, capable to deal with the conflict between knowledge and select high-quality knowledge for KG sharing.

In addition, the quality of historically submitted knowledge from different contributors should be dynamically evaluated and traced. In many cases, in order to ensure high quality KGs, the knowledge provided by contributors may be erroneously rejected in the absence of contrary evidence, especially for contributors with low credibility [16]. As the KG grows, so does the basis of quality judgment. Therefore, a dynamic assessment of the quality of new and historical knowledge is necessary.

Furthermore, the quality of the past knowledge provided by a contributor could be considered as a proof of work quality, which could be used to evaluate credibility [17]. For example, consistently delivering high-quality knowledge could make contributors progressively more trusted. The KQ assessment of the contribution history of a contributor can thus be used to access its credibility.

For the management of the sharing process, blockchain, as a decentralized distributed architecture, provides an automatically executed digital agreement among multiple collaborators to meet the verification, traceability and sharing of information [18], [19]. The smart contract of the blockchain facilitates implementation of the KQ management strategy in the sharing process, thereby ensuring that the knowledge provided by multiple contributors is published and shared according to the established policies. Moreover, its traceability facilitates the use of historical information to evaluate a contributor's credibility.

This paper proposes a blockchain-based KG collaborative and sharing framework (Fig. 1), which utilizes distributed ledgers to integrate and manage knowledge, including storage, query, update, knowledge filtering and sharing. Moreover, a dynamic KQ evaluation algorithm using multi-contributor knowledge consistency is designed and integrated into smart contract. It traces the knowledge submitted by different contributors through distributed ledgers, and employs the consistency between knowledge to design an iterative quality evaluation algorithm to dynamically update the knowledge quality. In this way, conflicts in knowledge are resolved, and high-quality knowledge is automatically selected. Meanwhile, an automatic contributor

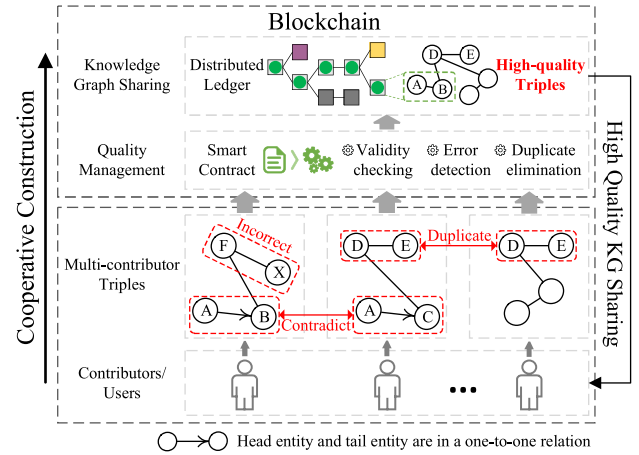


Fig. 1. Collaborative construction and sharing of high-quality KGs based on blockchain.

credibility management method is designed, which evaluates the credibility of contributors by tracing the historical records of contributors, and then restricts the knowledge release of low-quality contributors.

The contributions of this paper are as follows:

- A collaborative KG construction and sharing framework based on blockchain is proposed for the management of multi-contributor knowledge.
- A dynamic quality management method for multi-contributor knowledge is integrated into the proposed framework to improve knowledge quality during KG construction and sharing.
- A contributor credibility management method is introduced into the framework to limit the negative impact of malicious contributors.

The rest of this paper is organized as follows. After introducing the related work of this paper in Section II, the proposed framework and the KQ evaluation algorithm embedded in it are described in Sections III and IV, respectively. The detailed implementation of each component are introduced in Section V. The evaluation and discussion of the framework are presented in Section VI. Finally, Section VII concludes the paper.

## II. RELATED WORKS

### A. Motivation and Quality Security Assurance Requirements

Constructing KGs, which aggregate knowledge from the real world, often require substantial human efforts [20]. Therefore, many KGs are built through the collaboration of multiple contributors. However, the submission of erroneous knowledge is unavoidable, necessitating the control of quality within Knowledge Graphs [21].

The dimensions for evaluating knowledge quality are diverse [9], with most methods highly dependent on the collection of a substantial amount of additional information [22], [23], [24]. However, these methods prove challenging to implement in the construction of KGs that involve collaboration from

multiple contributors. In reality, the submission of substantial repetitive knowledge and the prevalence of redundancy are common in tasks involving collaborative construction by multiple contributors [25], [26]. That is, knowledge describing the same fact may be submitted multiple times by different contributors. Many methods exploit the consistency among these knowledge to assess its quality. This consistency refers to the degree of non-contradiction between different knowledge describing the same fact. The essence of this evaluation method is to use the agreement degree from other knowledge towards a specific knowledge as evidence of its high credibility to assess its quality, similar to ensemble algorithm [27]. This method does not rely on the collection of any additional information and only uses the knowledge submitted by contributors. This significantly facilitates the assurance of KG quality and simplifies the complexity of collaborative construction of knowledge graphs.

The security of a collaborative KG construction system can be bifurcated into two categories: data security and KG quality security. This classification is based on whether the objective is to prevent external malicious attacks on the framework or to guard against malicious submissions from within the framework. First, regarding the quality security of KGs, contributors might engage in certain behaviors within the permissible scope of the system's upload mechanism during its application process, leading to a potential compromise in KG quality. For example, contributors might submit incorrect knowledge in an attempt to replace correct triples, or republish shared knowledge in a bid to deceive the system and obtain higher credibility level<sup>2</sup>. Ensuring the quality security of the KG is a prerequisite to guarantee that the constructed KG can support its downstream applications. Therefore, this paper designs a rigorous process and quality evaluation mechanism to ensure the quality security of the KG.

Second, in terms of system data security, shared systems might be susceptible to external malicious attacks, leading to data forgery, loss, or tampering, which are common and fundamental protections requirements in most data storage systems. To address this issue, this paper leverages blockchain technology to ensure data privacy and security.

### B. Blockchain-Based KG Frameworks

Leveraging the advancements in distributed systems, blockchain technology has found extensive applications in crowdsourcing systems and cloud computing [33], [34]. Blockchain, serving as a distributed ledger fortified by encryption technology to safeguard data security, has seen broad adoption across various domains [35]. The blockchain packages transactions in blocks and each full node maintains a copy of the complete blockchain to ensure the security and integrity of transactions. As a digital agreement deployed on the blockchain, a smart contract realizes the formulation and automatic execution of business governance rules, and has a high degree of reliability and availability. Currently, there exist many blockchain frameworks designed to adapt to different application scenarios, which

TABLE I  
COMPARISON OF CURRENT FRAMEWORKS FOR MULTI-CONTRIBUTOR KG MANAGEMENT

| Sec. | Framework                    | Shareability | Security | Efficiency | Quality |
|------|------------------------------|--------------|----------|------------|---------|
| 2.1  | [22]                         | ○            | ○        | ○          | ●       |
| 2.1  | [23]                         | ○            | ○        | ●          | ○       |
| 2.1  | [24], [28], [29], [30], [31] | ○            | ○        | ●          | ●       |
| 2.1  | [32]                         | ●            | ○        | ○          | ●       |
| 2.2  | [36], [37], [38]             | ●            | ◐        | ○          | ○       |
| 2.2  | [39], [40]                   | ●            | ◐        | ◐          | ○       |
| 2.2  | [41]                         | ●            | ◐        | ○          | ●       |
| 2.2  | [42]                         | ●            | ◐        | ◐          | ●       |

\*○, ◐, and ● represent the three levels of the corresponding indicators of the framework from small to large. Specially, ◐ in the 'Security' column means that the framework only defends illegal modification of the shared KG, without concerning attack that revises shared KGs by publishing legal triples.

can be mainly divided into public, private and permissioned blockchains. Representative frameworks include Hyperledger, Ethereum MultiChain, etc. The blockchain architecture in this paper is based on Hyperledger, a private and permissioned blockchain framework.

Due to the advantages of blockchain in ensuring data privacy and security, many works attempt to combine KG with blockchain [36]. For example, Yao et al. [37] implements knowledge building of electronic health records through blockchain to achieve data security. Similarly, Joshi et al. [38] integrates the knowledge graph and blockchain to protect data security and shareability. In addition, as the amount of knowledge increases, the efficiency of the knowledge sharing system also needs to be optimized. Some works focus on the efficiency of the framework. Wang et al. [39] uses the distributed file system to preprocess the process files for building knowledge graphs to improve storage efficiency. Naim et al. [40] enhanced blockchain systems by adding a data graph service-layer, streamlining KG access, query, interlink, and retrieval. Regardless of the degree of efficiency optimization, a unifying characteristic of these studies is their emphasis on employing blockchain technology to ensure basic data security in shareable conditions. They neglect the importance of safeguarding the quality security of KGs. In the collaborative construction of shareable KG, contributors may submit incorrect knowledge unintentionally. Furthermore, some contributors may maliciously exploit potential omissions or mechanisms in the framework to undermine knowledge security, causing the knowledge shared by the system to not meet expectations, thereby posing risks to the application of KG.

Although with careful design, some frameworks can provide knowledge with high quality, such as the OpenKG chain framework in [41]. Besides, Wang et al. [42] implements a parallel healthcare system based on blockchain to achieve comprehensive medical data sharing with auditability and efficiency. However, they does not implement knowledge quality management, nor does it integrate methods to deal with malicious attacks into the knowledge quality management system.

To highlight the differences among existing works more clearly, this paper summarizes the focal points of the aforementioned studies in Table I. As can be seen, blockchain-based KG sharing system are not as efficient as traditional ones, as they prioritize ensuring shareability and safety. However, KGs

<sup>2</sup>The specific definition of a malicious contributor is described in Section VI-A.



TABLE II  
SYMBOLS USED TO REPRESENT DIFFERENT DATA TYPES AND OPERATIONS

|           | Symbol            | Description                                             |
|-----------|-------------------|---------------------------------------------------------|
| Data Type | $P\_TIME$         | The submission time of a triple                         |
|           | $E\_ID/R\_ID$     | The ID number of enti-                                  |
|           | $F\_ID/C\_ID$     | ties/relations/facts/contributors                       |
|           | $E\_NAME/R\_NAME$ | The name of entities/relations                          |
|           | $Q$               | The estimated quality of a triple                       |
|           | $F\_FLAG$         | The flag indicating whether a triple is a follower      |
|           | $PT\_NUM$         | The number of submitted triples of a contributor        |
|           | $CRD$             | The credibility of a contributor                        |
|           | $LS$              | The top level key-value store of the ledger state       |
|           | $(A, B, \dots)$   | Tuple consisting of data in types of A, B, ...          |
| Operation | $\{A\}$           | Set consisting of data in the type of A or with value A |
|           | $A-B$             | Data (type A) is the key of data (type B), one to one   |
|           | $A-\{B\}$         | Data (type A) is the key of data (type B), one to many  |
|           | $S[K]$            | The value of the key K in the key-value store S         |
|           | $T.A$             | The value of data type A of the tuple T                 |
|           | $S[K] = V$        | set $S[K]$ to value $V$                                 |
|           | $T.A = V$         | set $T.A$ to value $V$                                  |
|           | $A \cup B$        | Union of Set A and Set B                                |
|           | $A \setminus B$   | Remove all the elements of Set B from Set A             |
|           |                   |                                                         |

under existing blockchain frameworks still lack quality safeguards. Incorrect or low-quality knowledge may be submitted, and malicious knowledge contributors could potentially increase risks. In response to these issues, this paper aims to address both the quality and data security of sharable KG framework within an acceptable efficiency.

### III. FRAMEWORK ARCHITECTURE

To facilitate the use and quality management of KGs, we propose a trusted framework for KG sharing by many users and KG co-construction by many contributors simultaneously. The proposed framework introduces a 3-level knowledge storage approach into the consensus system to support trusted knowledge sharing and maintenance. Modular algorithms for contributor credibility assessment and multi-contributor KQ evaluation are designed to keep the triples in KG “state-of-the-art” (SOTA).<sup>3</sup>

To elaborate on the proposed framework, the rest of this section is organized as follows. Section III-A introduces the main workflow and basic mechanisms of the framework. Section III-B discusses the ledger design for the triple-level knowledge and the storage of contributor property supporting the subsequent KQ evaluation. Section III-C describes the smart contract design for KG quality maintenance. The symbols denoting the related concepts and operations are presented in Table II.

#### A. Overview

In this section, we provide an overview of important components from the aspect of smart contract, ledger storage, and consensus mechanism, see Fig. 2.

1) *Transaction Initiation & Smart Contract*: To start submitting a new triple, the corresponding contributor peer initiates a transaction through generating a ledger record. Next, the transaction is delivered to endorsement peers, which execute the operations of quality evaluation of the new triple together with corresponding triples from other contributors related to the

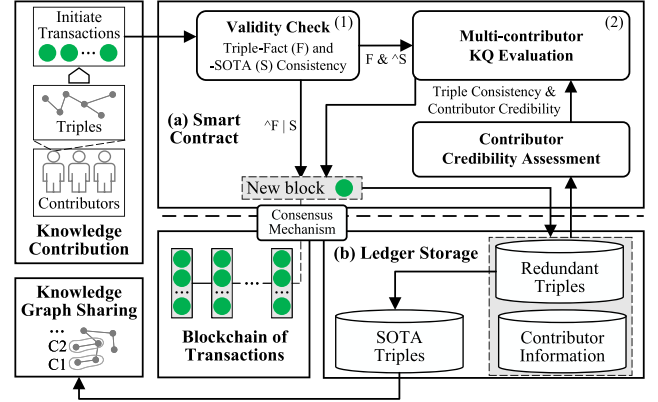


Fig. 2. Overview diagram of the proposed framework. (a) Smart Contract handles the uploaded triples and update them along with estimated quality to the ledger. It first (1) checks (F) whether the triple is related to existing fact or not, and (S) whether it is consistent with the SOTA triple. If (F) does not hold, a new fact is created. If (S) holds, the follower triple is recorded. If neither, (2) KQ evaluation is processed to estimated the triple inconsistent with the SOTA, and record the evaluated quality. (b) The ledger stores all the contributed triples along with information of contributors and quality. The highest-quality triple related to a fact is taken as the SOTA and shared.

same fact. Finally, the triple with the highest quality (i.e., SOTA triple) describing this fact is updated as a new “SOTA triple” and inserted into the KG. Related ledger states are also updated, such as the history of submitting knowledge of the contributor and the quality of other triples related to the same fact.

2) *Ledger Storage*: We organize three data structures of SOTA triples, redundant triples, and contributor information in the ledger storage. SOTA triples are high-quality knowledge shared to contributors, redundant triples record all triples uploaded by each contributor, and contributor information record IDs and credibility of contributors.

Triples in the KG are in basic form of “head entity-relation-tail entity”, and a sequence of the entity and relation identification numbers is employed for triple storage. The submitted triples describe various facts where different triples could be both related to one or different facts. For KG storage, triples are organized and stored according to the facts they describe. The facts are numbered to group the triples from multi-contributors, which provides a basis for the triple quality evaluation depending on the consistency comparison among triples related to the same fact. Moreover, triples submitted by different contributors are also classified and stored according to different facts, which facilitates the triple update when a contributor changes the statement on certain facts and supports the contributor credibility assessment process.

3) *Consensus Mechanism*: The security and integrity of our framework are underpinned by the blockchain, which ensures that once published, triples cannot be maliciously altered. This guarantee is provided by the underlying consensus mechanism. However, it’s essential to note that while the consensus mechanism ensures the immutability of published triples, it does not inherently prevent the addition of new, yet meaningless triples, such as those introduced by a *spammer*. To counteract such attacks, our framework incorporates a quality evaluation

<sup>3</sup>“SOTA triple” refers to the triple with the best quality describing a fact.

algorithm complemented by specific anti-attack mechanisms, see Section III-C.

*Workflow of the Consensus Mechanism:* 1) *Triple Submission*: All submitted triples, associated facts, and knowledge submission transactions are stored on a distributed ledger. Each contributor (or peer) maintains a copy of this ledger. 2) *Transaction Initiation*: When a new triple is introduced, the corresponding contributor peer initiates a transaction. 3) *Pre-Preparation and Preparation*: The primary node broadcasts the transaction to backup nodes. Backup nodes then acknowledge the receipt of the transaction and send a preparation message to all nodes. 4) *Commitment*: Nodes exchange commitment messages based on the transaction. Once a node receives a 2/3 majority of commitment messages, it moves to the next phase. 5) *Block Validation*: The created blocks are dispatched to all peers for validation. This step ensures that the transactions within the block remain unaltered. During this phase, each transaction within a block is marked as either valid or invalid. 6) *Ledger Update*: Validated blocks are appended to each peer's local ledger. The state of the ledger is then updated based on the valid transactions within the block.

This comprehensive process ensures consensus across the entire transaction flow, synchronizing a sequence of transactions across all peers. In essence, it guarantees full-circle verification of transaction correctness, facilitating trusted knowledge publishing and effective KG quality management.

## B. Ledger Design

This part introduces the underlying ledger design supporting the full-cycle of KG sharing and maintenance from three aspects, i.e., the KG storage, redundancy and contributor properties. We use the following notation to describe the design of the data structure. “ $A-B$ ” means that  $A$  is the key of  $B$  in a “one-to-one” relation, where as “ $A-\{B\}$ ” refers to “one-to-many” relation.

1) *KG Storage*: To store the KG and its quality management information on the ledger, the triples (i.e., entities and relations) are transformed into three data structures manipulated separately. These data are maintained in a key-value store as part of the ledger state as follows:

- Entities:  $E\_ID-E\_NAME$
- Relations:  $R\_ID-R\_NAME$
- SOTA Triples:  $F\_ID-(E\_ID, R\_ID, E\_ID)$
- All Triples:  $F\_ID-\{C\_ID\}$

where each entity and relation is identified with a unique ID number. On this basis, each triple is stored in the form of ordered ID numbers corresponding to its head entity, relation and tail entity. For example, a triple  $(en1, re1, en2)$  is stored as a list of “ $ID_{e1}, ID_{r1}, ID_{e2}$ ” which correspond to the ID number of entity  $en1$ , relation  $re1$  and entity  $en2$ . Since a knowledge may be contributed by many contributors, the triples with the best quality among them are selected as “SOTA Triples” to construct the KG. Moreover, all the triples are organized based on their related facts, i.e.,  $F\_ID$ . By this way, triples from different contributors (identified by  $C\_ID$ ) could be gathered into groups, as shown by the key-value store of “All Triples”. The grouped storage of triples is conducive to count the consistency

among knowledge describing a fact and evaluate the credibility of contributors.

The triples in the same group could be referred to through the contributor state introduced in Part C-(3). These triples participate in the knowledge quality evaluation when a new triple is submitted to that group. In other words, the organization of multi-contributor triples based on  $FACT\_ID$  supports the quality evaluation process. For example, three contributors  $C_1, C_2, C_3$  contribute three triples related to a fact  $F_1$ , respectively. Among them the triple  $(en2, re2, en3)$  from  $C_2$  is evaluated as the highest quality one to be the SOTA triple. Then “SOTA Triples” has a corresponding key-value item:  $F_1-(ID_{e2}, ID_{r2}, ID_{e3})$ . The corresponding key-value item in “All Triples” is:  $F_1-\{C_1, C_2, C_3\}$ .

2) *Redundancy Storage*: Four redundant key-value structures are maintained to facilitate knowledge retrieval based on the query of entity or relation, which could be presented as:

- Entity to ID:  $E\_NAME-E\_ID$
- Relation to ID:  $R\_NAME-R\_ID$
- Entity to Triple:  $E\_ID-\{F\_ID\}$
- Relation to Triple:  $R\_ID-\{F\_ID\}$

When a contributor provides a new triple related to an existing fact, the  $F\_ID$  could be queried through the entity or relation of the knowledge based on the above redundancy structures. Also the redundancy could support partial KG query by users thus enhancing the usability of the shared KG.

If a contributor submits a knowledge about certain non-existing fact, a new  $F\_ID$  is created. Admittedly, this mechanism may increase the risk of malicious attacks on the system, such as persistently posting spam triples and requesting new  $F\_ID$ . Therefore, this paper introduces a garbage recycling mechanism in Section III-C to defend against this attack.

3) *Contributor Property Storage*: The properties of contributors are maintained in the ledger state to support the quality evaluation of triples and SOTA triple update. Specifically, the knowledge provided by each contributor on each fact is stored together with the corresponding estimated quality. The key-value structure of contributor stored on the ledger is represented as:

- Contributor:  $C\_ID-\{F\_ID-(E\_ID, R\_ID, E\_ID, Q)\}$

where the “ $Q$ ” represents the estimated quality of the corresponding triple, and the key-value set of each contributor ( $C\_ID$ ) is updated when the contributor submits a new triple. Based on this contributor property storage mechanism, it is convenient to evaluate the overall credibility of each contributor by the quality of its submitted knowledge. This overall credibility is in turn used in the quality evaluation of triples related to any fact submitted by the contributor, which is introduced in Section III-C.

## C. Smart Contract for KG Quality Maintenance

This part introduces the smart contract that conducts the validity check of knowledge, contributor credibility assessment and knowledge quality evaluation. In these processes, several security mechanisms are integrated to resist various malicious attacks.

1) *Knowledge Validity Check*: The validity of a new submitted triple on an existing fact is checked by two procedures. First, the ID numbers of the fact, entities and relation should exist in the corresponding key-value store. Besides, whether the submitted triple is related to the fact should be verified according to certain criteria.

In addition, a garbage  $F\_ID$  recycling mechanism is designed to check the triples on new facts. A new  $F\_ID$  is assigned to such triple when submitted. In the subsequent certain time span, if more than a certain threshold amount of triples related to that  $F\_ID$  is submitted by other contributors, this  $F\_ID$  is forever valid or it is marked suspicious. The suspicious facts with their related triples are delivered to the peers pre-approved based on their reputation and reliability, voting for validity. The  $F\_ID$  without enough votes is removed from the key-value store and recycled for other triples on new facts.

The configuration of the time span can be established to be proportional to the average submission interval of existing knowledge with a parameter  $\alpha$ . This setup uses historical submission information to estimate regular knowledge update intervals. If the time interval between subsequent submission of a specific knowledge surpass this set time span, it could be interpreted as unattended and suspicious. The parameter  $\alpha$  is set as 10 in this research and can be adjusted in accordance with specific situation [43]. Consequently, a larger  $\alpha$  indicates that submitted knowledge would be accepted more comprehensively.

2) *Contributor Credibility Assessment*: The credibility of each contributor is assessed by calculating the average quality of its containing triples. A “follower constraint” strategy is introduced to promise the fairness of this process. When a contributor submits a new triple on certain fact, its quality is evaluated in different ways depending on whether it repeats with the shared knowledge (i.e the SOTA triple). If it does, the triple is marked as “follower”, whose quality is directly assigned with a decreased value of the quality of SOTA triple:

$$Q = \lambda Q_{ST} \quad (1)$$

where  $\lambda$  is the decreasing factor and set as 0.5 in the following experiments.  $Q, Q_{ST}$  represent the quality of submitted and SOTA triple, respectively. If not, the quality of triples related to the same fact is re-evaluated and the quality of only non-follower triples is updated.

With this strategy, “follower” triples are limited in improving the credibility of contributors by setting them to a lower and non-increasable quality. The strategy takes the repeated SOTA triples submitting as a non-contribution option which does not benefit the contributor credibility. In fact, only the disagreement on the SOTA triple could possibly contribute to correct the error triples and improve the KG quality. Therefore, the non-follower triples get a full score of quality during each re-evaluation process. It's worth noting that each triple is marked “follower” or not only at the time of publication. In other words, the contribution of a triple is determined. A “follower” triple never turns to a non-follower one even if the SOTA triple is changed and vice versa. This mechanism could prevent contributors from maliciously increasing their credibility by uploading shared knowledge repeatedly.

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**Algorithm 1: EvaluateKQ: Multi-Contributor KQ Evaluation.**


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**Input:** Knowledge on same fact  $k_1, k_2, \dots, k_N$ ;  
Credibility of contributors  $p_1, p_2, \dots, p_N$ ;  
Number of total iterations  $L$ .  
**Output:** Knowledge quality  $q_1, q_2, \dots, q_N$ .

```

1 for  $j=1:N$  do
2    $q_j^0 = p_j$ ; // Initialization
3 end
4 for  $l=1:L$  do
5   // In the  $l^{\text{th}}$  iteration
6   for  $i=1:N$  do
7     // Quality update of  $k_i$ 
8     for  $\forall e \neq i$  do
9       // Validate  $k_i$  with  $k_e$ 
10      if  $k_e = k_i$  then
11         $W_{i,e}^{l-1} = \frac{1}{l}(1 - q_i^{l-1})$ ; // Eq. (4)
12      else
13         $W_{i,e}^{l-1} = -\frac{1}{l}q_i^{l-1}$ ; // Eq. (4)
14      end
15       $Z_{i,e}^{l-1} = \frac{1}{2}(q_e^{l-1} - q_i^{l-1} + 1)q_e^{l-1}$ ; // Eq. (5)
16    end
17     $q_i^l = q_i^{l-1} + \frac{1}{N} \sum_{e \neq i} W_{i,e}^{l-1} \times Z_{i,e}^{l-1}$ ; // Eq. (3)
18  end
19 end
20 for  $j=1:N$  do
21    $q_j = q_j^L$ ;
22 end
23 return  $q_1, q_2, \dots, q_N$ ;

```

---

3) *Knowledge Quality Evaluation*: A multi-contributor knowledge quality evaluation algorithm is designed to evaluate the triple quality from different contributors and integrated into the smart contract as a module. This module takes 1) the overall credibility of each contributor, and 2) the triples related to a common  $F\_ID$  as input and produces the quality of each triple. This process occurs when a different triple from the SOTA one is submitted and evaluates the quality of triples in the corresponding  $F\_ID$  group. Note that only the quality of non-follower triples are updated with the evaluation results. The triple with the highest quality is taken as the new SOTA triple. This algorithm is subsequently introduced in Section IV.

#### IV. MULTI-CONTRIBUTOR KNOWLEDGE QUALITY EVALUATION ALGORITHM

In order to ensure the quality of knowledge in the KG, we propose a knowledge (triple) quality evaluation method based on the consistency of knowledge. It evaluates the quality of knowledge by constructing a unified evaluation indicator and selects high-quality triples from conflicting descriptions of facts.

##### A. Symbol Definition

Let  $N$  denote the number of input triples describing a fact. These triples are denoted as  $\{k_1, k_2, \dots, k_N\}$  and the input overall credibility of the corresponding contributors is denoted by  $\{p_1, p_2, \dots, p_N\}$ . An iterative algorithm is designed to evaluate the quality of these triples. In an iteration, each triple takes turn to be the investigated triple  $k_i$  and the other triples are taken as external triples  $k_e$ . The quality of  $k_i$  is updated by validation using each  $k_e$ , and the quality of all triples is updated by taking



each triple as the investigated one in turn. The estimated quality of the two types of triples are expressed as  $q_i^l$  and  $q_e^l$  in the  $l^{\text{th}}$  iteration.

### B. Algorithm Introduction

For the investigated knowledge  $k_i$  and its external knowledge  $k_e$ , their quality is initialized ( $l = 0$ ) with the corresponding contributor credibility, presented as:

$$q_i^0 = p_i, q_e^0 = p_e \quad e \in \{1, 2, \dots, N\}, e \neq i \quad (2)$$

The motivation for the updated process comes from the mutual authentication of knowledge. When the investigated knowledge is agreed or disagreed by other knowledge, its quality rises or falls. The total update amount consists of the total evaluation of investigated knowledge by other knowledge. Specifically, the iterative update in the  $l^{\text{th}}$  ( $l \in \mathbb{N}^+$ ) iteration for the quality of  $k_i$  is shown as:

$$q_i^l = q_i^{l-1} + \frac{1}{N} \sum_{e \neq i} W_{i,e}^{l-1} \times Z_{i,e}^{l-1} \quad (3)$$

where  $W_{i,e}^{l-1}$  is the upper bound of the quality update and  $Z_{i,e}^{l-1}$  is the update degree. The update bound  $W_{i,e}^{l-1}$  indicates the extreme value that knowledge quality could reach, and reflects the situation when knowledge is extremely supported or denied. In addition, it is related to the number of iterations, representing that the knowledge quality tends to be stable after a certain number of iterations.  $W_{i,e}^{l-1}$  is defined as:

$$W_{i,e}^{l-1} = \begin{cases} \frac{1}{l}(1 - q_i^{l-1}) & k_e = k_i \\ -\frac{1}{l}q_i^{l-1} & k_e \neq k_i \end{cases} \quad (4)$$

where  $k_e = k_i$  means that  $k_i$  and  $k_e$  have identical head entities, relations and tail entities ( $k_e$  agrees  $k_i$ ), and  $k_e \neq k_i$  represents that at least one item is different ( $k_e$  disagrees  $k_i$ ). It shows that the quality of  $k_i$  increases if  $k_i$  is agreed by certain  $k_e$  and decreases if not.  $\frac{1}{l}$  is a convergence factor, which balances the impact of the external comparison and empirical (contributor) credibility on the quality evaluation.

The increase/decrease degree  $Z_{i,e}^{l-1}$  depends on the current quality of  $k_i$  and  $k_e$ . When the external knowledge is authoritative and the gap between the two is large, the quality of the investigated knowledge could be greatly updated. Concretely, the definition of  $Z_{i,e}^{l-1}$  is expressed as:

$$Z_{i,e}^{l-1} = \frac{1}{2}(q_e^{l-1} - q_i^{l-1} + 1)q_e^{l-1} \quad (5)$$

where  $q_i^{l-1}, q_e^{l-1}$  could be considered as a prior quality of the knowledge to the quality update process of  $l^{\text{th}}$  iteration.  $Z_{i,e}^{l-1}$  is jointly determined by the priors of both external and investigated knowledge, consisting of two factors,  $q_e^{l-1}$  and  $\frac{1}{2}(q_e^{l-1} - q_i^{l-1} + 1)$ . For the  $q_e^{l-1}$ , it is individually utilized as a factor of the update degree, since the external knowledge  $k_e$  with higher prior is supposed to be more convincing. As for the other factor, it includes the prior of  $k_i$  except for that of  $k_e$ . If  $q_i^{l-1}$  is low, the external knowledge with high prior quality could affect the quality update of  $k_i$  to a great extent, since it is a much more

credible knowledge than  $k_i$ . On the contrary, if the prior of  $k_i$  is high, that  $k_e$  does not impact on the quality update of  $k_i$  as much as the former case, since the two knowledge are both empirically credible and more confident on their statements. The overall algorithm is presented in Algorithm 1.

### C. Security and Time Complexity of Knowledge Quality Evaluation

This section demonstrates from a theoretical point of view that the proposed algorithm meets the requirements of the task in terms of security and efficiency. The practical verification can refer to the experimental part.

1) *Time complexity*: When the proposed algorithm assesses the quality of knowledge, it needs to compare each knowledge with other knowledge describing the same fact. Assuming that there are  $n$  knowledge for a fact, the quality update of each knowledge needs to be compared  $n - 1$  times, so  $l$  iterations of the quality of these knowledge need to compare  $nl(n - 1)$  times. The number of calculations required for each comparison, as shown in (4) and (5), is a constant. Therefore, the time complexity of knowledge quality assessment is  $O(n^2 l)$ .

In the real world,  $n$  is usually much smaller than the total number of knowledge  $M$ , since typically only a few transactions are submitted on the same fact. Moreover, the quality evaluation only occurs when a new triple disagrees with the SOTA triple. Therefore, the time consumed on executing the smart contract for a transaction is given by:

$$(1 - \mu)O(1) + \mu O(n^2 l) * \frac{M}{n} = O(nlM\mu) \quad (6)$$

where  $\mu$  is the proportion of KQ evaluation transactions and  $\frac{M}{n}$  is an estimate of the number of facts, using  $n$ , the number of knowledge corresponding to a fact and  $M$  the total number of knowledge. This shows that the time complexity spent is approximately linear, so the proposed quality assessment algorithm is efficient.

2) *Security*: The proposed algorithm addresses spammers among the three security issues raised in Section II-A by evaluating the quality of each contributor and each piece of knowledge. Under this system, in order for the knowledge submitted to have wide-ranging impact, the quality of that knowledge must be rated highly. It can be seen from (3), (4), and (5) that high quality comes from the approval of other knowledge. Therefore, if the knowledge contributed by a spammer is inconsistent with many other knowledge, the knowledge cannot be highly evaluated, and thus cannot have a great voice. In this case, the personal credibility of spammers will also be greatly reduced, which in turn affects their subsequent submissions of knowledge.

## V. IMPLEMENTATION

This section provides a deep dive into the foundational elements of our proposed framework. We elucidate the architecture of our permissioned blockchain, detailing the intricacies of ledger storage and the implementation of our smart contract. Furthermore, we have highlighted a mechanism specifically designed to accelerate the triple uploading process in real-world

TABLE III

USED SYMBOLS IN THE ILLUSTRATION OF THE PROCEDURES IMPLEMENTING THE SMART CONTRACT

| Symbol           | Description                                                                                |
|------------------|--------------------------------------------------------------------------------------------|
| $nT$             | The triple to be submitted by a contributor                                                |
| $ID_h/ID_r/ID_t$ | The IDs representing the head entity/relation/tail entity of $nT$                          |
| $ID_c$           | The C_ID of the contributor that submits $nT$                                              |
| $ID_f$           | The F_ID that $nT$ is related to                                                           |
| TIME             | The time when $nT$ is submitted                                                            |
| $sT$             | The SOTA triple related to $ID_f$                                                          |
| IDGroup          | The group of C_ID of contributors ever submitting triples related to $ID_f$                |
| $s$              | The C_ID in IDGroup (element of IDGroup)                                                   |
| TGroup           | The group of submitted triples related to $ID_f$ corresponding to IDGroup                  |
| $t_s$            | The triple by the contributor with C_ID $s$ in IDGroup (element of TGroup)                 |
| Creds            | The group of credibility of contributors corresponding to IDGroup                          |
| NewCreds         | The group of new credibility of contributors corresponding to IDGroup                      |
| $cr_s$           | The credibility of the contributor with C_ID $s$ in IDGroup (element of Creds or NewCreds) |
| QGroup           | The group of evaluated quality of triples corresponding to TGroup                          |
| $q_s$            | The evaluated quality of triple $t_s$ in TGroup (element of QGroup)                        |
| $\lambda$        | The decreasing factor for calculating the quality of follower triples.                     |
| $q_0$            | The initial quality of a triple related to a new fact.                                     |

TABLE IV  
STRUCTURE OF THE LEDGER STATE

| First Level  | Second Level       | Key-Value Store                                                             | Abbr |
|--------------|--------------------|-----------------------------------------------------------------------------|------|
| Triples      | Entities           | $\{E\_ID-E\_NAME\}$                                                         | ID2E |
|              | Relations          | $\{R\_ID-R\_NAME\}$                                                         | ID2R |
|              | SOTA Triples       | $\{F\_ID-(E\_ID,R\_ID,E\_ID,Q)\}$                                           | ST   |
|              | All Triples        | $\{F\_ID-\{C\_ID\}\}$                                                       | AT   |
| Contributors | -                  | $\{C\_ID-(CRD,PT\_NUM,Pt\{F\_ID-(E\_ID,R\_ID,E\_ID,Q,P\_TIME,F\_FLAG)\})\}$ | C    |
| Redundancy   | Entity to ID       | $\{E\_NAME-E\_ID\}$                                                         | E2ID |
|              | Relation to ID     | $\{R\_NAME-R\_ID\}$                                                         | R2ID |
|              | Entity to Triple   | $\{E\_ID-\{F\_ID\}\}$                                                       | E2T  |
|              | Relation to Triple | $\{R\_ID-\{F\_ID\}\}$                                                       | R2T  |

"LS" is the top level. The righter columns are the more detailed level.

applications. For ease of reference and clarity, associated symbols integral to our framework are cataloged in Tables II and III.

#### A. Implementation of Key-Value Store of Ledger State

The main structure of key-value stores representing the ledger state has been introduced in Section III-B, including the state of provided triples, contributors and redundancy. For implementation, additional information is included to facilitate the operation of the smart contract level. The detailed implementation of the structure of the ledger state (LS) is presented in Table IV.

Main changes compared to Section III-B are the state of "SOTA Triples" and "Contributors". For "SOTA Triples", the quality of each SOTA triple in the KG is also stored. The purpose is to facilitate the quality setting of the "follower" triple, whose quality is unchangeable and is set to lower than that of the SOTA triple. To identify these "follower" triples, a flag type " $F\_FLAG$ " is added to the structure of "Contributors". In addition, the current credibility " $CRD$ " and the number of submitted triples " $PT\_NUM$ " are supplemented to speed up the calculation of contributor credibility. Finally, the submitting time " $P\_TIME$ " of

#### Procedure 1: TrPublish: Publishing A New Triple.

**Input:** Triple to be submitted  $h$  (head entity),  $r$  (relation),  $t$  (tail entity); Contributor ID  $ID_c$ .

**Output:** Transaction to update the ledger state.

```

1  $ID_h, ID_r, ID_t \leftarrow E2ID[h], R2ID[r], E2ID[t]$ ;
2 if  $ID_h/ID_r/ID_t$  is NULL then
3    $ID_h/ID_r/ID_t \leftarrow$  Create new E_ID/R_ID;
4    $ID_f \leftarrow$  Create new F_ID;
5    $trans0 \leftarrow \{Update\ of\ ID2E/ID2R/E2ID/R2ID\}$ ;
6 else
7    $Temp \leftarrow \{E2T[ID_h], R2T[ID_r], E2T[ID_t]\}$ ;
8    $Candidate \leftarrow \cup\{X \cap Y | X, Y \in Temp, X \neq Y\}$ ;
9    $ID_f \leftarrow choose\_from(Candidate)$ ; // Executed by
   the contributor
10   $ID_f \leftarrow$  Create new F_ID if  $ID_f$  is NULL;
11 end
12  $N \leftarrow C[ID_c].PT\_NUM, D \leftarrow C[ID_c].CRD$ ;
13 if  $ID_f$  is new then
14    $trans1 \leftarrow trans0 \cup \{ST[ID_f] = (ID_h, ID_r, ID_t, q_0),$ 
      $AT[ID_f] \cup = \{ID_c\}, C[ID_c].PT\_NUM = N+1,$ 
      $C[ID_c].CRD = \frac{N \times D + q_0}{N+1}, C[ID_c].Pt[ID_f] =$ 
      $(ID_h, ID_r, ID_t, q_0, TIME, FALSE)\}$ ;
15   PutOnChain( $trans1$ );
16   return NULL;
17 end
18 if  $ID_c \in AT[ID_f]$  then
   // Ever submitted by the contributor
19    $q \leftarrow C[ID_c].Pt[ID_f].Q$ ;
20    $trans2 \leftarrow trans0 \cup \{C[ID_c].PT\_NUM = N-1, C[ID_c].CRD$ 
      $= \frac{N \times D - q}{N-1}, C[ID_c].Pt[ID_f] =$ 
      $(ID_h, ID_r, ID_t, /, TIME, /)\}$ ;
21 else
   // First submitted by the contributor
22    $trans2 \leftarrow trans0 \cup \{AT[ID_f] \cup = \{ID_c\}, C[ID_c].Pt[ID_f]$ 
      $= (ID_h, ID_r, ID_t, /, TIME, /)\}$ ;
23 end
24 return  $trans2$ ;
```

each triple is added to defense malicious attack. If a contributor repeats submitting triples related to a fact in a short time, it is termed as the malicious attack and forbidden.

All the required information during the workflow could be obtained by querying the ledger state. For example, the group of triples related to certain  $FACT\_ID$  "F0001" could be queried as follows:

- $\{s_1, s_2, \dots, s_n\} \leftarrow LS[Triples][All\ triples][F0001]$
- $t_i \leftarrow LS[Contributors][s_i].Pt[F0001], i = 1, 2, \dots, n$

where the group of contributors ever submitting triples related to "F0001" is queried in the first step. The group of triples related to "F0001" is queried through the properties of these contributors.

#### B. Implementation of Smart Contract

In this part, detailed implementations of the smart contract are introduced, as shown in Procedures 1, 2 and 3. Note that we use abbreviations  $*.T$  to represent  $(*.E\_ID, *.R\_ID, *.E\_ID)$ , the triple in  $*$ . The included data flow and control flow are illustrated by the framework architecture in Fig. 3 for a more clear view. Note that the variables  $trans1, \dots, trans7$  in the procedures correspond to the graphic elements "Trans1", ..., "Trans7" in Fig. 3.

##### Triple Upload (Procedure 1)

This procedure accepts the head, tail entity and relation submitted by the contributor, and performs certain pre-processing.



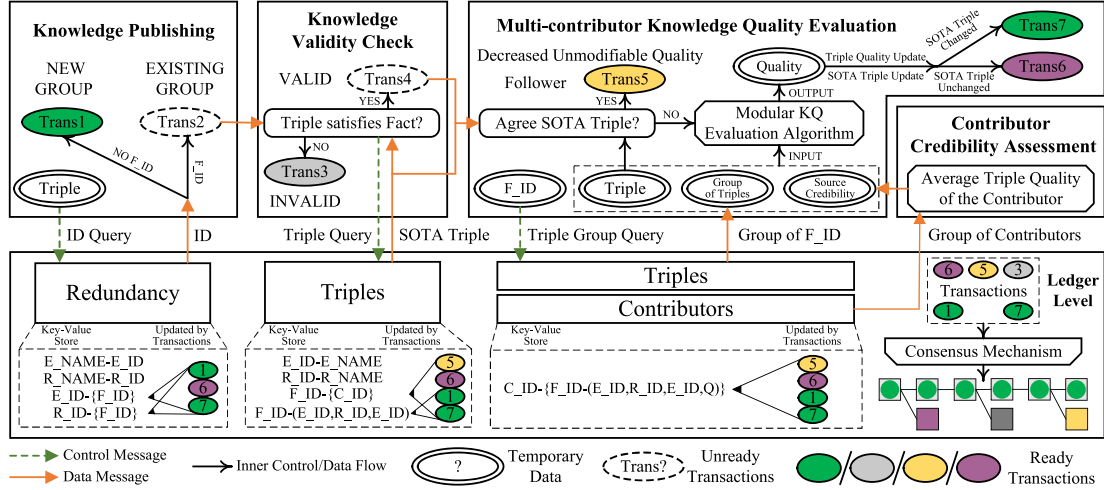


Fig. 3. Detailed workflow diagram of the proposed framework. The figure element “Consensus Mechanism” refers to the endorsing, ordering and validating phase of the blockchain technique. The figure element “Modular KQ Evaluation Algorithm” is introduced in Section IV.

### Procedure 2: CheckVF: Checking Validity & Follower Property of the Triple.

**Input:** IDs representing the new triple  $ID_h, ID_r, ID_t$ ;  
Fact that the triple is related to  $ID_f$ ;  
Contributor ID  $ID_c$ ; Previous transaction  $trans2$ .  
**Output:** Transaction to update the ledger state.

```

1  $nT \leftarrow (ID_h, ID_r, ID_t)$ ;
2  $sT \leftarrow ST[ID_f].T$ ;
3 if  $nT$  is not related to  $sT$  then
4    $trans3 \leftarrow Invalid(trans2)$ ;
5    $PutOnChain(trans3)$ ;
6   return NULL;
7 end
8  $trans4 \leftarrow Valid(trans2)$ ;
9 if  $nT$  is identical with  $sT$  then
10   $q \leftarrow ST[ID_f].Q$ ;
11   $trans5 \leftarrow trans4 \cup \{C[ID_c].Pt[ID_f].Q = \lambda q,$ 
12     $C[ID_c].Pt[ID_f].F\_FLAG = TRUE\}$ ;
13   $PutOnChain(trans5)$ ;
14  return NULL;
15 end
16  $trans4 \leftarrow trans4 \cup \{C[ID_c].Pt[ID_f].F\_FLAG = FALSE\}$ ;
17 return  $trans4$ ;

```

Specifically, Procedure 1 first judges whether a new fact needs to be created according to the submitted triple, or it is classified as an existing fact in the knowledge graph.

#### Validity & Follower Check (Procedure 2)

This procedure checks whether the uploaded triple is consistent with its fact, meanwhile preventing cheaters from improving their own credibility in unintended ways to avoid future harm. After the validity check (lines 1-8), if the submitted triple is consistent with the current SOTA triple, then the triple will not undergo subsequent quality evaluation, but will be directly assigned a cut value of the quality of the SOTA triple. Under this strategy, even if cheaters keep submitting SOTA triples, they will be evaluated as having low credibility.

#### Quality Evaluation (Procedure 3)

This procedure evaluates and updates the triples in the same group as the new triple (same  $F\_ID$ ) if the new triple is valid and not a “follower”. Specially, Procedure 3 query the group of triples and corresponding contributors related to  $ID_f$  first, Then

### Procedure 3: UpdateKQ: Evaluating and Updating the Quality of Triples Related to $ID_f$ .

**Input:** IDs representing the new triple  $ID_h, ID_r, ID_t$ ;  
Fact that the triple is related to  $ID_f$ ;  
Contributor ID  $ID_c$ ; Previous transaction  $trans4$ .  
**Output:** Transaction to update the ledger state.

```

1  $nT \leftarrow (ID_h, ID_r, ID_t)$ ;
2  $sT \leftarrow ST[ID_f]$ ;
3  $IDGroup \leftarrow AT[ID_f]$ ;
4  $TGroup \leftarrow \{C[s][ID_f].T | s \in IDGroup\}$ ;
5  $Creds \leftarrow \{C[s].CRD | s \in IDGroup\}$ ;
6  $QGroup \leftarrow EvaluateKQ(TGroup, Creds)$ ;
7  $maxQ, sT \leftarrow \max_{t_s} q_s, \max_{t_s} q_s \in QGroup$ ;
8  $QGroup \leftarrow \{q_s = 1 \text{ if } t_s = sT \text{ else } q_s = 0 \text{ for } t_s \in TGroup\}$ ;
9  $NewCreds \leftarrow \{cr_s | cr_s = \frac{C[s].CRD \times C[s].PT\_NUM - C[s].Pt[ID_f].Q + q_s}{C[s].PT\_NUM}, s \in IDGroup, q_s \in QGroup\}$ ;
10  $trans6 \leftarrow trans4 \cup \{C[s][ID_f].Q = q_s, C[s].CRD = cr_s | s \in IDGroup, q_s \in QGroup, cr_s \in NewCreds, C[s][ID_f].F\_FLAG \text{ is FALSE}\}$ ;
11 if  $sT$  is changed then
12   $trans7 \leftarrow trans6 \cup \{ST[ID_f] = (sT, maxQ)\}$ ;
13   $PutOnChain(trans7)$ ;
14 else
15   $PutOnChain(trans6)$ ;
16 end

```

evaluates the quality of the triples using Algorithm 1. Further, use the obtained results to update the quality of the relevant triples, the SOTA triples and the credibility of the contributors.

## VI. EVALUATION AND DISCUSSION

In this section, a qualitative discussion of the utility and security is first provided in Section VI-A. Experimental results to show the practical performance of the proposed framework are provided in Section VI-B.

### A. Utility and Security

Blockchain technology has been widely utilized to support crypto currency applications such as Bitcoin and Ethereum. In this scenario, transactions are continuously and frequently

created, and the security of digital assets is guaranteed by the cooperation of smart contracts and underlying consensus mechanism. In comparison, the KG sharing application has much less frequent transaction creation and simpler security requirements. The proposed framework satisfies the utility and security requirements of KG sharing applications as follows:

*Utility : 1) Time complexity:* The total time spent in proposed framework mainly comes from knowledge quality assessment and publishing transactions. The time complexity of knowledge quality assessment, as discussed in Section IV-B, is approximately linearly related to the amount of knowledge in the KG. The number of transaction releases is greatly reduced under the batch contribution mechanism. Therefore, this framework is efficient.

*2) Extensibility:* The additional attributes on entities, relations and triples could be simply extended to the store structure. To store entity attributes, the key-value structure of entities could be extended by adding a structure “*E\_ID-{ATT\_TYPE-ATT\_CONTENT}*” to the ledger state. The attributes of entities could be available for both users and the quality evaluation process. For example, in a patient information sharing scenario, critical attributes such as the age and gender could be included in the proposed framework. These attributes are visible to all the participated medical institutions for treat research purpose. Moreover, incorrect collected data could be detected by rule design based on the existing attributes (quality evaluation process) to maintain a high-quality KG.

*Security: 1) Data Security:* Our framework ensures data security using the blockchain, which safeguards against malicious alterations of published triples, backed by a consensus mechanism and underlying data structure.

Each peer has a complete copy of the ledger and blockchain data. Any block that a peer attempts to add will be verified by other peers to prevent data tampering. At the same time, each new block contains the hash value of the previous block, forming a chain structure. This means that modifying the information of a certain block without leaving any trace requires modifying all subsequent blocks, which is very difficult.

*2) Knowledge Quality Security:* The proposed framework uses various strategies to prevent threats to knowledge security from unintentional errors or intentional attacks by contributors. Unintentionally committed erroneous knowledge is detected with quality assessment. While for malicious contributors, we classify them according to their characteristics and discuss them separately:

- Sybil: This malicious contributor creates vast amounts of identities to flood the network with meaningless transactions or by submitting false triples.
- Cheater: This malicious contributors attempts to exploit vulnerabilities in the framework to break knowledge security. In our proposed framework, a “cheater” is characterized as an entity that duplicates SOTA triples (highest quality triple) and re-publishes them to get high credibility.
- Spammer: This malicious contributors submits incorrect knowledge to the system in an attempt to replace the correct triple.

The proposed framework prevents Sybil attacks by employing a permissioned blockchain, where only trusted nodes are authorized to submit knowledge, while malicious nodes are isolated. Moreover, the frequency of submitting triples is limited so as to prevent transaction flooding.

“Follower” detection mechanism is designed to prevent cheater attack. Newly submitted triples that are identical with the SOTA ones are marked as “follower” triples, whose quality is fixed at a lower value and never changed. Hence, the contributor is evaluated as being of low credibility even if they keep copying SOTA triples.

For spammer, the proposed framework defends against it through the module of contributor credibility assessment. The triples of contributors who submit casually will be of low quality because no other triples agree with them. Its own credibility will also be close to 0, thus being largely ignored in the subsequent evaluation process.

## B. Performance Evaluation

This section assesses the framework’s ability to maintain correct triples, defend against malicious attacks, and its feasibility in real-world KGs.

For the first two experiments, we use FB13 triples [44] from the test data. This provides 23,733 pairs of correct and incorrect triples, sharing the same head entity and relation but different tail entities. Multi-contributors are simulated, each submitting correct FB13 triples with a probability  $p$ , indicating varying credibility levels. Correct triples serve as ground truth for assessing output SOTA triples’ accuracy. For experiment 3), we use Freebase’s FB2M sub-dataset [45] to assess the framework’s utility for large-scale KGs. This dataset includes over 2 M entities and 5 k relationships. Multiple contributors submit these triples to our framework, and we evaluate the transaction processing time consumed.

*1) Accuracy of SOTA triples:* Two existing consensus algorithms utilized in blockchain to evaluate the triple quality are adopted to compared with the proposed algorithm. They are the algorithm in [46] (MKPB) and the weighted majority consensus algorithm (WMCA) in [47]. MKPB treats every contributor equally. When contributors disagree, the MKPB simply assumes that the side with the numerical advantage is right. WMCA is an improvement on MKPB, which takes the credibility of contributors as the weight and believes in highly credible contributors. However, this practice requires a good understanding of the credibility of contributors. Accurate credibility assessment is difficult when contributors submit small amounts of knowledge.

The experiments simulate contributors submitting some triples. Assume that the triple of these submissions is limited to facts corresponding to 100 pairs of triples in FB13. To assess the effect of number of contributor on method performance, we conduct experiments with varying numbers of contributors. Each contributor’s credibility, derived from a randomly generated value  $p$  in the range 0 to 1, is held consistent across all experiments to prevent credibility distribution influence. In practice, we create a group of contributors (25 in our experiment), each

TABLE V  
ACCURACY OF SOTA TRIPLES OF THE PROPOSED FRAMEWORK WITH DIFFERENT NUMBER OF SPAMMERS

| T_NUM \ SP_NUM | 0     | 2     | 4     | 6     | 8     | 10    | 12    | 14    | 16    | 18    | Coefficient of Variation | Maximum Credibility |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------------|---------------------|
| 100            | 0.636 | 0.589 | 0.581 | 0.518 | 0.499 | 0.445 | 0.405 | 0.402 | 0.387 | 0.380 | 0.194                    | 0.82                |
| 600            | 0.659 | 0.652 | 0.623 | 0.586 | 0.543 | 0.559 | 0.473 | 0.477 | 0.464 | 0.434 | 0.150                    | 0.72                |
| 1100           | 0.767 | 0.761 | 0.738 | 0.685 | 0.683 | 0.695 | 0.617 | 0.606 | 0.606 | 0.566 | 0.105                    | 0.33                |
| 1600           | 0.853 | 0.829 | 0.816 | 0.796 | 0.793 | 0.789 | 0.731 | 0.728 | 0.713 | 0.716 | 0.065                    | 0.16                |
| 2100           | 0.888 | 0.873 | 0.877 | 0.857 | 0.851 | 0.829 | 0.802 | 0.822 | 0.805 | 0.791 | 0.041                    | 0.11                |
| 2600           | 0.912 | 0.907 | 0.898 | 0.899 | 0.883 | 0.859 | 0.855 | 0.855 | 0.852 | 0.848 | 0.029                    | 0.11                |
| 3100           | 0.933 | 0.920 | 0.918 | 0.919 | 0.913 | 0.887 | 0.893 | 0.888 | 0.877 | 0.880 | 0.022                    | 0.10                |
| 3600           | 0.938 | 0.931 | 0.928 | 0.939 | 0.918 | 0.902 | 0.923 | 0.908 | 0.890 | 0.897 | 0.019                    | 0.10                |
| 4100           | 0.951 | 0.942 | 0.937 | 0.950 | 0.931 | 0.917 | 0.924 | 0.923 | 0.903 | 0.910 | 0.017                    | 0.10                |
| 4600           | 0.948 | 0.948 | 0.942 | 0.955 | 0.943 | 0.925 | 0.931 | 0.931 | 0.918 | 0.913 | 0.015                    | 0.09                |
| CPU time (/ms) | 1520  | 1762  | 2032  | 2254  | 2455  | 2634  | 2898  | 3088  | 3328  | 3343  | —                        | —                   |

\*T\_NUM is the total number of triples submitted by contributors. SP\_NUM is the number of spammers. The coefficient of variation is the ratio of the standard deviation to the mean, which represents the degree of data volatility. Maximum credibility is the maximum of spammer credibility.

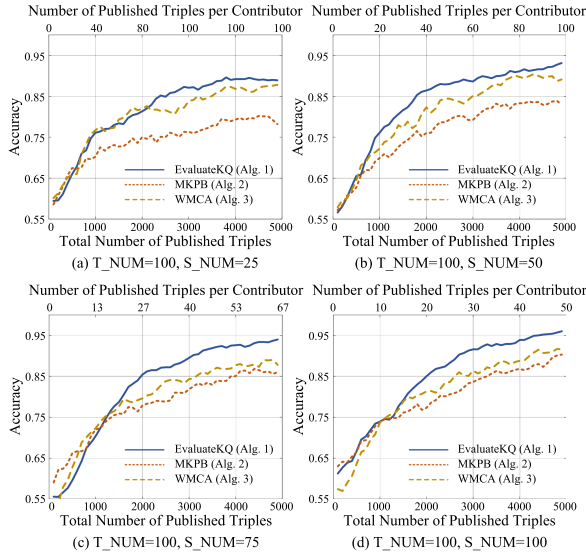


Fig. 4. Comparison of three different KQ evaluation algorithms integrated with the proposed framework. T\_NUM is the number of triples the KG contains. S\_NUM is the number of contributors participating in the KG sharing process.

with an assigned credibility, then replicate each contributor to increase the numbers, preserving the distribution. In this manner, we expanded the number of contributors (S\_NUM) from 25 to 50, 75, and 100. A total of 5,000 triples are submitted. The experiment with each set of parameters is repeated 10 times, and the results of average accuracy are presented in Fig. 4.

Fig. 4 demonstrates that Algorithm 1 surpasses Algorithm 2 and 3 with large submitted triples. This underscores the efficiency of Algorithm 1 and 3 in utilizing contributor credibility for more precise evaluations than Algorithm 2. Despite sharing this utilization, the proposed method outdoes Algorithm 3 because of its integrated computation relying on triple pair consistency for quality estimation. However, Algorithm 2 excels with fewer triples per contributor, probably due to credibility inaccuracies with fewer submissions. Unaffected by these inaccuracies, Algorithm 2's performance decreases as submission numbers rise and credibility assessments improve, with Algorithm 1 and 3 subsequently improving.

As the number of contributors increases, the accuracy of SOTA triples increases (from about 87% in Fig. 4(a) to about 96% in Fig. 4(d)). This conclusion may not be rigorous enough,

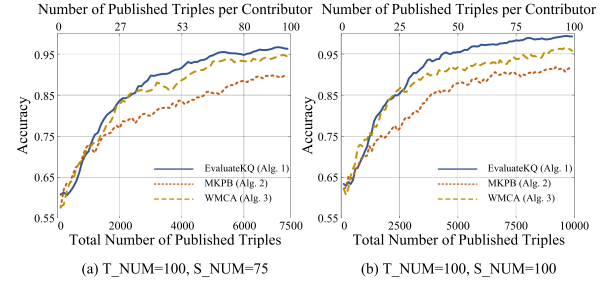


Fig. 5. Accuracy results of the SOTA triples in different numbers of contributors under the same number condition of submitted triples per contributor.

as the experiment held the total number of triples constant, resulting in a change in the number of triples per contributor when the number of contributors varied. Therefore, a supplementary experiment was conducted to show the performance with 75 and 100 contributors where the number of submitted triples per contributor is the same as 25 contributors. The results, shown in Fig. 5, illustrate that the accuracy of SOTA triples is significantly improved (by comparing Fig. 4(a), (b), 5(a) and (b)) as the number of contributors increases.

2) *Against spammer attack:* In our experiments, we simulated additional spammers alongside normal contributors. These spammers attempted to attack the system by submitting triples with random tail entities of each fact. With 200 triples in the KG and 20 normal contributors of random credibility, the number of spammers varied from 0 to 18. Results are in Table V. The coefficient of variation (C.V.) of accuracy under varying spammers is reported. This showed that spammers negatively impact the accuracy of SOTA triples when submitted triples (T\_NUM) are few. However, as T\_NUM grows, C.V. decreases significantly, reducing the spammers' damage. This is due to the algorithm correctly identifying spam triples as low quality when submissions increase. Consequently, spammers' credibility decreases, reducing their influence on the evaluation process as seen in Table V.

Note that the accuracy under the condition of more spammers is lower even if the submitted triples are sufficient. As the number of spammers (SP\_NUM) increases, the accuracy decreases under the condition that T\_NUM=4600. Note that the maximum of the spammer credibility is already degraded to 0.09, hardly able to harm the evaluation process. In fact, this decrease of accuracy is caused by the reduced number of



TABLE VI  
CONSUMED TIME OF CONSTRUCTING AND MAINTAINING FB2M ON THE  
BLOCKCHAIN

| Number of triples<br>to publish | Triple number<br>per transaction | Consumed<br>time (s) | Time per<br>triple (s) |
|---------------------------------|----------------------------------|----------------------|------------------------|
| 10M<br>(construction)           | 4000                             | 10986                | 1.09e-03               |
|                                 | 3000                             | 17964                | 1.79e-03               |
|                                 | 2000                             | 28212                | 2.82e-03               |
|                                 | 1000                             | 58306                | 5.83e-03               |
| 10k (maintenance)               | 100                              | 441                  | 4.41e-02               |

triples submitted by normal contributors relative to spammers. Since the total number of triples is identical in all cases, the number of triples submitted by normal contributors decreases as the number of spammers increases. This explanation is consistent with the experimental results. For example, the accuracy of case “T\_NUM=2600, SP\_NUM=0” is close to that of case “T\_NUM=4600, SP\_NUM=18”. The number of triples submitted by normal contributors in case “T\_NUM=4600, SP\_NUM=18” is approximately  $2,421 (4600 / (20 + 18) * 20)$ , which is close to 2,600. The CPU time for submitting 4,600 triples is also reported in Table V, which is an important criteria to estimate the utility of the proposed framework in practice. This time is purely consumed by Procedures 1–3, representing the execution efficiency of the designed smart contract for KG management. It shows that the consumed time increases approximately linearly with the number of contributors which is quite efficient to deal with the KG management.

3) *Feasibility in real-world KGs*: To evaluate the feasibility of the proposed framework in practice, a blockchain is built by utilizing hyperledger fabric. Three nodes are simulated to participate in the knowledge contributing of the dataset FB2M. A total of 2 M entities, 7 k relations and 10 M triples are published on the blockchain. The experiments are conducted on three dedicated servers with Intel Core i5-8600 k CPU@3.60GHz×6 and 500 GB SSD, running Ubuntu 20.04.4 LTS.

The transaction speed of the constructed blockchain is about 10 transactions per second when the smart contract is implemented without any real operation. The consumed time is unacceptable for uploading 10 M triples if one transaction is used for one triple. Therefore, the smart contract is designed to process multi-triple contributions in one transaction. Thousands of triples could be submitted at one time, processed in sequence and published on the blockchain as one transaction. With this approach, the main factor of consumed time, publishing transactions, is greatly reduced in frequency. For example, to publish 1,000 triples needs to publish 1,000 transactions, causing at least 100 seconds. If the 1,000 triples are processed in an transaction, only one transaction is published thus greatly reducing the consumed time of the consensus procedure in the blockchain. In practice, thousands of triples or more are usually published once, where one transaction for multiple triples is suitable and efficient. Different sizes of triples in a transaction are set and the corresponding time for uploading 10 M triples and maintaining this KG are shown in Table VI.

The results in the Table show that the consumed time for uploading 10 M triples decreases as the number of triples in one transaction increases. The total time of constructing FB2M

could be reduced to three hours when there are 4,000 triples in one transaction. Note that the time per triple is longer when maintaining the KG than constructing it, since the number of entities and relations is higher than that in the construction process, thus leading to more time querying the fact a new triple is related to.

## VII. CONCLUSION

A blockchain-based high-quality KG collaborative construction and sharing framework to address the conflicts of multi-contributor knowledge and to maintain a high-quality KG was proposed. The framework uses a blockchain adopted to a data structure feasible for the storage of triples. Three main procedures (Procedures 1–3) are designed into the smart contract to implement the KG sharing and quality control. In addition, a modular knowledge quality evaluation algorithm and a follower detection mechanism are proposed to ensure the fairness and security of the framework. These mechanisms are also effective against malicious attacks such as spammers and cheaters. Moreover, the time complexity is acceptable for practical applications and the framework has good extensibility to include more detailed properties of triples. Experiments are used to verify that the proposed framework is effective to maintain high-quality KGs and has strong capability to defend against malicious attacks.

For future work, due to the fact that the utilized blockchain framework hyperledger fabric is short in the query of information on the ledger, where formal query statements are not supported. To facilitate the further utility of the proposed framework, we are working on a query language integrated in the smart contract to query the triples conveniently and efficiently. In addition, sparsely submitted knowledge may be greatly affected by the credibility of contributors, so we consider using large language models to provide a more reliable basis for the initialization of knowledge [48].

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**Xiangyu Wang** received the BSc degree from Donghua University, Shanghai, China, and the PhD degree in data science from the University of Science and Technology of China (USTC). He is currently an associate researcher with the School of Computer Science and Technology, USTC. His research interests include knowledge engineering and machine learning.



**Taiyu Ban** received the BSc degree in computer science and technology from the School of the Gifted Young, University of Science and Technology of China, Hefei, China, in 2020. He is currently working toward the PhD degree with the School of Computer Science and Technology, University of Science and Technology of China, Hefei, China. His current research interests include machine learning and knowledge engineering.



**Lyuzhou Chen** received the BSc degree from the University of Science and Technology of China, Hefei, China, in 2020. He is currently working toward the PhD degree with the School of Data Science, University of Science and Technology of China, Hefei, China. His current research interests include ensemble learning, knowledge engineering and causal learning.



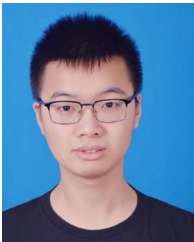
**Muhammad Usman** received the BSc degree from International Islamic University, Islamabad, Pakistan, in 2006, master's degree from SZABIST, Islamabad, Pakistan, in 2017. He is currently working toward the PhD degree in computer science with the School of Computer Science and Technology, USTC, Hefei, China. His recent research interests include Data Stream Mining, Machine Learning and Knowledge Discovery.



**Huanhuan Chen** (Senior Member, IEEE) received the BSc degree from the University of Science and Technology of China (USTC), Hefei, China, and the PhD degree in computer science from the University of Birmingham, Birmingham, U.K. He is currently a full professor with the School of Computer Science and Technology, USTC. His current research interests include neural networks, Bayesian inference, and evolutionary computation.



**Yifeng Guan** received the BSc degree from the University of Science and Technology of China, Hefei, China, in 2021. He is currently working toward the MSc degree with the School of Computer Science and Technology, University of Science and Technology of China, Hefei, China. His current research interests include knowledge engineering and blockchain.



**Derui Lyu** received the BSc degree in intelligence science and technology from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2021. He is currently working toward the PhD degree with the School of Computer Science and Technology, University of Science and Technology of China, Hefei, China. His current research interests include distributed system and knowledge engineering.



**Jian Cheng** received the BSc degree in automation, the MSc degree in control theory and control engineering, and the PhD degree in communication and information system from the China University of Mining and Technology, Xuzhou, China, in 1997, 2003, and 2008, respectively. He is currently a professor and the chief scientist with the Central Research Institute, China Coal Research Institute, Beijing, China. His research interests include machine learning and pattern recognition.



Canada.

**Cyril Leung** (Life Member, IEEE) received the BSc degree (Hons.) from the Imperial College, University of London, UK, and the MS and PhD degrees in electrical engineering from Stanford University. He served as the associate dean of Research and Graduate Studies with the Faculty of Applied Science, UBC, from 2008 to 2011. His research interests include wireless communication systems, data security, and technologies to support ageless aging for the elderly. He is a member of the association of professional Engineers and Geoscientists of British Columbia,



create novel experiences and dimensions in game design, interactive narrative, and other real world agent systems.

**Chunyan Miao** (Fellow, IEEE) received the BS degree from Shandong University, Jinan, China, in 1988, and the MS and PhD degrees from Nanyang Technological University (NTU), Singapore, in 1998 and 2003, respectively. She is currently a professor with the School of Computer Science and Engineering, NTU, and the director of the Joint NTU-UBC Research Centre of Excellence in Active Living for the Elderly (LILY). Her research interests include infusing intelligent agents into interactive new media (virtual, mixed, mobile, and pervasive media) to